Electronics Basics

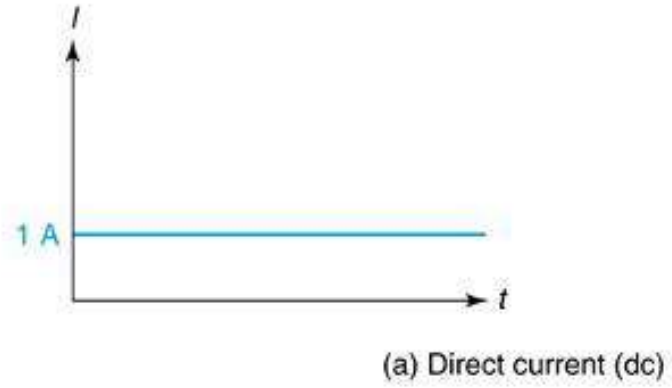
AC, Capacitor, Inductor

Electricity: DC and AC

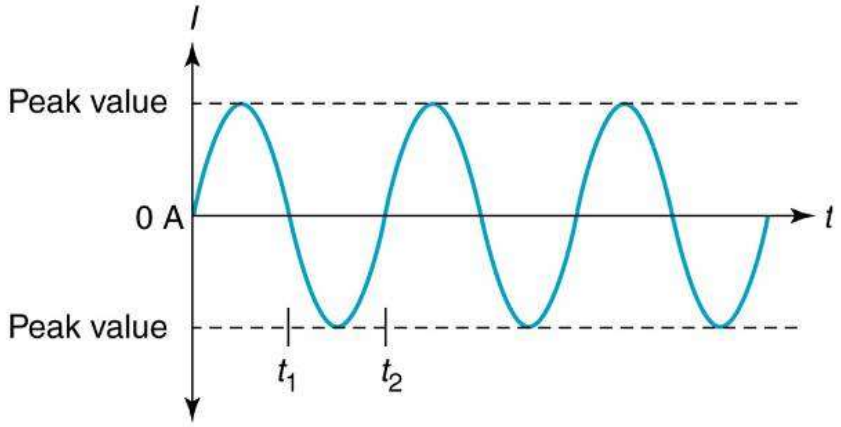
- **Direct Current (DC) – Unidirectional**, always flows in **one direction**:

Ex: current from **battery**

- **Alternating Current (AC)** – Bidirectional, periodically **changes direction**.
  Ex: **power outlet**



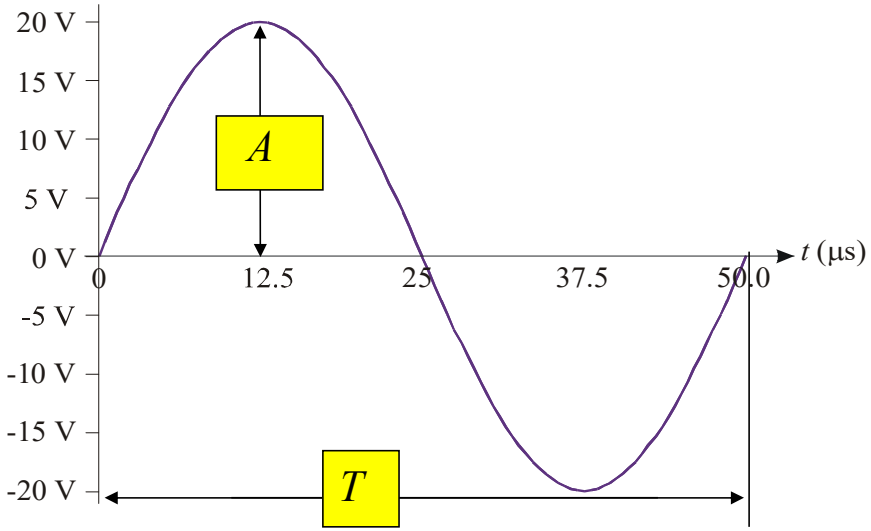
(a) Direct current (dc)



Amplitude, Period and Frequency

- The **amplitude** is the maximum value of a voltage or current measured from the mean;
- **Period** is the time interval for one complete cycle.

- **Frequency (*f*)** is the number of cycles that a sine wave completes in one second.
- Frequency is measured in **hertz** (Hz).



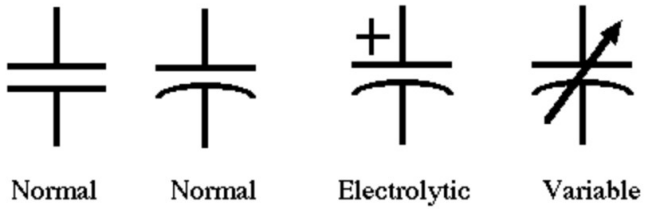
The amplitude (A) = 20 V

The period is 50.0 μs

$$f = \frac{1}{T} = \frac{1}{50\ \mu s} = 20\ \text{kHz}$$

Capacitor

- ***Capacitance:*** The **ability** of a component to **store energy** in the form of an **electrostatic charge**
  - Unit of Measure – **Farad (F)**
- **Capacitor** – is a component designed to provide a specific measure of **capacitance**

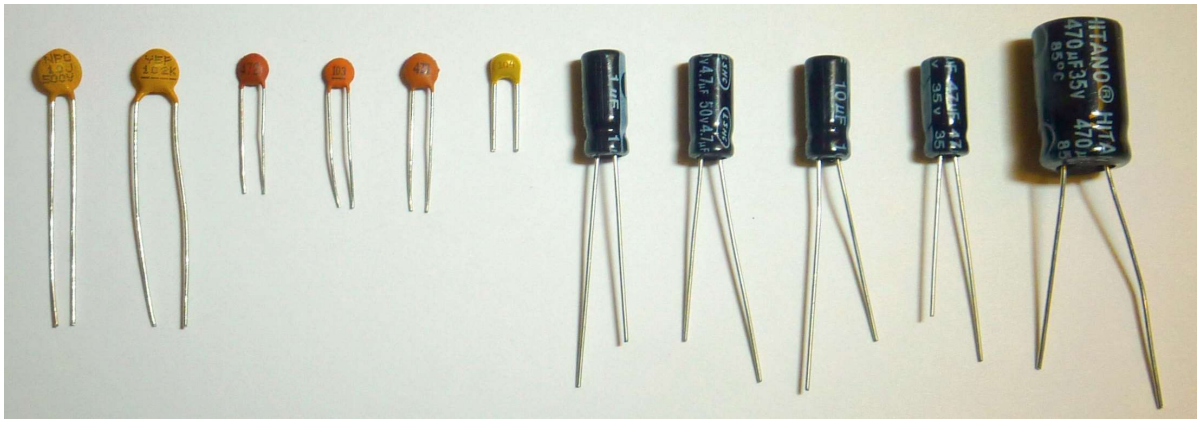






Types of Capacitors

- **Non- polarized Capacitors:**
  - Orientation Doesn't Matter: These capacitors can be connected in either direction.
  - Lower (picofarads to microfarads)
  - e.g: **Ceramic capacitors**
- **Polarized Capacitors**
  - Must be connected with correct polarity
  - High (usually in microfarads to farads)
  - e.g: **Electrolytic Capacitors**



Capacitor Labeling

- **Two-digit code**: the number represents the value of the component in pF.
  - Example: **15** means 15 pF
- **Three-digit code**: the code is interpreted like the first three digits of a resistor code.
  - **473** = 47 x 10^3 pF = 47 nF ; **104** = 100,000 pF = 0.1 µF

    Note: The numbers 6 and 7 are not used as multiplier values
    The numbers 8 and 9 are decoded as follows: 8 = 0.01 and 9 = 0.1;
    Example: 158 = 0.15 pF
- **Voltage Rating**: 50V indicates a capacitor can withstand 50 volts.

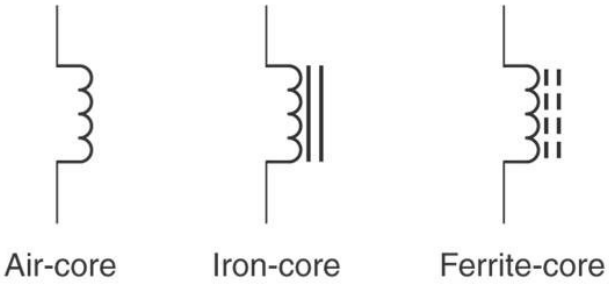


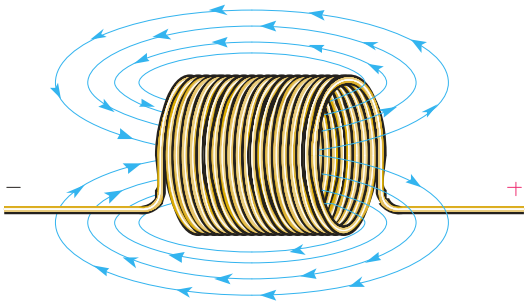




Inductor

- **An inductor** is a coil that stores energy in a magnetic field when electric current flows through it.
- **Unit** for inductance is – **Henry (H)**
- **Symbols:**

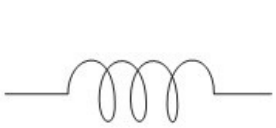




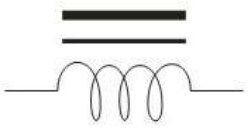
Current through the coil → creates a magnetic field.
If **current changes** → the coil generates a **voltage** that **opposes the change**.

Types of Inductors

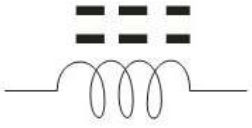
Common symbols for inductors (coils) are



Air core



Iron core



Ferrite core



Variable

Practical Inductors



Encapsulated



Wire wound, high current



Toroid coil



Variable

- Schematics often use the air core symbol regardless of the actual core material.
- The coil frequently is wound on a ferromagnetic core to increase the inductance.
- Toroids are a commonly used , very efficient inductor